

Multiple Choice (10 marks)

1. The inverse of $-i$ in the multiplicative group, $\{1, -1, i, -i\}$ is
 - (a) 1
 - (b) -1
 - (c) i
 - (d) $-i$
2. Find the order of the factor group $(\mathbb{Z}_{11} \times \mathbb{Z}_{15})/(\langle 1, 1 \rangle)$
 - (a) 1
 - (b) 2
 - (c) 5
 - (d) 11
3. The maximum number of acute angles in a convex 10-gon in the Euclidean plane is
 - (a) 1
 - (b) 2
 - (c) 3
 - (d) 4
4. Find the number of elements in the indicated cyclic group: The cyclic subgroup of $\mathbb{Z}_{30}$ generated by 25.
 - (a) 25
 - (b) 5
 - (c) 6
 - (d) 30
5. A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x ?
 - (a) $x^2/9$
 - (b) $x^2/8$
 - (c) $x^2/4$
 - (d) x^2

True / False (10 marks)

Answer True or False

6. True or False: There exist continuous functions from $\mathbb{R}$ to $\mathbb{Q}$ that are non-constant and maintain continuity across their entire domain.
7. True or False: Every free abelian group is torsion free, meaning it has no elements of finite order.
8. True or False: The equation $z^2 = |z|^2$ describes a pair of isolated points in the complex z -plane.
8. True or False: The probability that the player who makes the first toss wins the game is $1/4$.
9. True or False: Every integral domain with characteristic 0 is indeed infinite.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of the composition of functions on injectivity. If $g \circ f: S \rightarrow U$ is one-to-one, analyze why it must follow that f is also one-to-one.
12. Discuss the possible dimensions of the intersection of two 4-dimensional subspaces V and W within a 7-dimensional vector space X , and explain why certain dimensions cannot occur.

End of Paper